

Getting Started Guide - CompSoc Challenge - AdaHack 2026

Datasets and APIs

Satellite and imagery

- [Copernicus Data Space STAC API \(Sentinel-1/2/3/5P imagery\)](#)
- [Microsoft Planetary Computer STAC \(Landsat, Sentinel, MODIS, no login for many collections\)](#)
- [NASA FIRMS \(active fire and thermal anomaly detections\)](#)

Earth, water and air sensors

- [BGS Sensor Data API](#)
- [USGS Earthquake Catalog API \(baseline for seismic work\)](#)
- [EMSC seismic event feed](#)
- [Environment Agency Flood Monitoring API \(river levels, UK\)](#)
- [OpenAQ \(global air quality sensors\)](#)
- [Open-Meteo \(free weather and reanalysis, no API key\)](#)
- [Copernicus Climate Data Store \(ERA5 reanalysis\)](#)

Space weather

- [NOAA SWPC \(solar wind, Kp index, aurora forecasts\)](#)
- [INTERMAGNET \(magnetometer data\)](#)

Weather and general

- [US National Weather Service API](#)
- [NASA Open Datasets](#)
- [NASA Open APIs](#)
- [OpenStreetMap API](#)

Technologies

Git

We'd recommend using Git for managing your project. Git allows for version control and having multiple developers work on the same project. A common platform for Git is GitHub. If this is new to you, we'd recommend their [getting started guide](#). Another good resource is [Oh Shit, Git!?!](#).

Python

Python is a modern programming language which is designed to be understandable and beginner friendly.

Resources:

- If it's your first time, we'd recommend having a look at their [getting started guide](#).
- [Here's a guide on installing packages with pip](#)

Python Packages

- [NumPy](#) - arrays and fast maths, very common and what most of these other libraries build off of. Extensive library of common algorithms in its submodules.
- [Pandas](#) - useful for handling and transforming data, the industry standard for dataframes in Python.
- [Polars](#) - a modern, often faster replacement for Pandas.
- [SciPy](#) - submodules for modeling, statistics, signal processing, and more.
- [scikit-learn](#) - machine-learning and modeling
- [Matplotlib](#) - very common basic plotting library.
- [Seaborn](#) - more advanced statistical plots, built on top of Matplotlib.